

AN AI-DRIVEN HOTEL RESERVATION AND ROOM MANAGEMENT PLATFORM WITH DYNAMIC PRICING AND DEMAND FORECASTING

A Literature Survey and Comparative Review

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Abstract—The hospitality industry increasingly depends on rapid, data-driven decision-making to manage room inventory, set competitive prices, and predict guest demand across fluctuating booking conditions. Manual reservation handling and static pricing rules are progressively being replaced by artificial intelligence techniques capable of learning from historical booking patterns, occupancy trends, competitor rates, and guest behaviour. This literature survey reviews ten reference works related to AI-driven dynamic pricing, occupancy and demand forecasting, booking-cancellation prediction, conversational reservation assistants, and Internet-of-Things-enabled room management. The reviewed works indicate a common movement from fixed-rate, manually operated hotel systems toward adaptive platforms that price rooms in real time, anticipate cancellations and demand shifts, and automate guest-facing reservation and room-management tasks. The survey compares the principal focus, technological direction, and expected contribution of each reference paper and identifies gaps concerning integrated pricing-forecasting pipelines, explainability, network-wide revenue effects, and evaluation across diverse property types. Based on the reviewed literature, an integrated AI-Driven Hotel Reservation and Room Management Platform with Dynamic Pricing and Demand Forecasting is proposed as a focused research direction.

Keywords—Hotel reservation systems, artificial intelligence, machine learning, dynamic pricing, demand forecasting, room management, revenue management, cancellation prediction, chatbots, Internet of Things, smart hotels

I. INTRODUCTION

Hotels generate continuous streams of operational data through property management systems, online travel agencies, direct booking engines, guest review platforms, and in-room sensors. As competition among properties intensifies and guest expectations shift toward instant, personalised service, the traditional approach of setting fixed room rates and processing reservations manually becomes increasingly inadequate. Occupancy, seasonality, local events, competitor pricing, and last-minute cancellations interact in ways that static rate cards cannot capture, so a price that is appropriate on one day may leave revenue on the table or discourage bookings on another.

Artificial intelligence offers a pathway from reactive, manually adjusted hotel operations toward predictive and adaptive reservation and room-management platforms. Machine-learning

models can learn relationships between historical booking curves, pricing history, and realised occupancy, while classification models can flag reservations that are likely to be cancelled. Conversational agents can reduce the friction of the booking process, and Internet-of-Things (IoT) sensing can extend intelligent management into the physical room itself. The reviewed literature reflects this transition through work on deep-learning-based dynamic pricing, price-elasticity and occupancy modelling, fairness-aware discriminatory pricing, cancellation-forecasting classifiers, AI-driven reservation chatbots, IoT-enabled smart-room management, and personalised hotel recommendation [1]–[10].

The objective of this literature survey is to examine ten reference works relevant to the proposed AI-Driven Hotel Reservation and Room Management Platform with Dynamic Pricing and Demand Forecasting. The review focuses on the problem addressed by each work, its AI or computational direction as indicated by the paper's title, abstract, and publication details, and its relevance to dynamic pricing, demand forecasting, reservation handling, and room management. The survey also identifies common limitations and research opportunities that can guide the design of an integrated platform.

II. REVIEW METHODOLOGY

The review was structured around ten reference papers selected because their titles and abstracts directly address AI-driven dynamic pricing, occupancy and demand forecasting, booking-cancellation prediction, reservation chatbots, and IoT-enabled smart-hotel and room-management systems [1]–[9]. A personalised hotel-recommendation paper [10] was also included because recommendation-system concepts are relevant to the guest-facing selection stage of the proposed reservation platform.

For each reference, the review extracts only information supported by the publicly available abstract, bibliographic details, and, where accessible, the full text of the paper. Where a citation does not specify a dataset, evaluation metric, or detailed algorithmic configuration, the survey does not assign an unsupported value or claim. The papers are discussed thematically to identify how pricing, forecasting, cancellation management, conversational interfaces, and physical room intelligence can be combined into a unified hotel-management architecture.

III. LITERATURE SURVEY

A. Deep-Learning-Based Dynamic Pricing for Hotel Revenue Management

Zhang, Qiu, Wu, Wang, and Luo present a dynamic pricing system for hotel revenue management that first analyses a hotel and its competitors to establish a reasonable base price and then forecasts future occupancy using a sequence-learning model that integrates a Deep Factorization-Machine with a sequence-to-sequence architecture [1]. The work is closely aligned with the proposed problem because it directly links occupancy forecasting with the generation of a price recommendation, rather than treating pricing and forecasting as separate concerns.

An important implication of this work is that dynamic pricing should not be built on current occupancy alone; a forecasting component that anticipates near-future occupancy allows the pricing engine to adjust rates before demand fully materialises. This integrated forecasting-to-pricing perspective is central to the proposed platform.

B. Data-Analysis-Driven Dynamic Pricing and Revenue Optimisation

Gao investigates the optimisation of hotel revenue management through dynamic pricing algorithms combined with statistical data-analysis techniques, including ANOVA, correlation analysis, chi-square testing, and multiple linear regression applied to hotel pricing data [2]. The paper identifies hotel type, star rating, location, and review count as leading factors that influence room pricing, providing an evidence base for the features a dynamic-pricing model should consider.

For the proposed platform, this reference motivates a feature-engineering stage in which structured attributes of a property and its competitive set are incorporated alongside time-varying occupancy and demand signals before a price recommendation is generated.

C. Price-Elasticity Modelling for Occupancy Prediction

Zhu et al. propose a price-elasticity framework for occupancy prediction in hotel dynamic pricing that addresses the sparsity of room-level occupancy data by transferring information from hotel-level occupancy predictions and by explicitly modelling the endogenous relationship between price and demand [3]. This work is significant because it treats occupancy not merely as an outcome to be forecast but as a quantity that responds to the price itself, which is essential for a system that intends to set, rather than only predict around, room rates.

Within the proposed platform, this direction suggests that demand forecasting and price optimisation cannot be fully separated: the forecasting module should be capable of estimating how occupancy would change under alternative candidate prices before a final rate is recommended.

D. Fairness-Constrained Discriminatory Pricing with Reinforcement Learning

Wang, Xie, Jian, Liu, and Lv propose a two-stage deep-reinforcement-learning-driven dynamic discriminatory pricing

model for hotel rooms that unifies fixed pricing, dynamic pricing, discriminatory pricing, and dynamic discriminatory pricing within a single framework subject to fairness constraints [4]. The reinforcement-learning algorithm derives pricing schemes that satisfy both group and temporal fairness while remaining computationally efficient enough for near-real-time use.

This reference is relevant to the proposed system because it highlights that purely profit-maximising dynamic pricing can create perceptions of unfairness among guests; a practical reservation platform should therefore incorporate fairness or transparency constraints into its pricing-optimisation objective rather than optimising revenue alone.

E. Bayesian Modelling of Booking-Cancellation Behaviour

Jishan, Singh, Ghosh, Alam, Mahmud, and Paul apply Bayesian models to predict hotel booking cancellations, framing cancellation forecasting as a probabilistic problem that can incorporate prior knowledge and update predictions as new booking information becomes available [5]. The probabilistic framing is useful because it can express uncertainty in a cancellation forecast rather than issuing only a binary prediction.

For the proposed platform, this suggests that a demand-forecasting module should account for the probability that confirmed reservations will not materialise, allowing net-demand estimates—rather than raw booking counts—to feed into the pricing and room-allocation logic.

F. Machine-Learning Classification for Cancellation Prediction and Net-Demand Calculation

Antonio, de Almeida, and Nunes develop a machine-learning-based classification system, deployed in a production environment across two hotels, that trains a model daily on property-management-system data to identify bookings that are likely to cancel and to calculate net demand accordingly [6]. Their evaluation, supported by A/B testing, shows that proactively contacting guests identified as likely to cancel measurably reduces the cancellation rate.

This work is directly relevant to the proposed platform because it demonstrates that a cancellation-prediction model can be operationalised, not merely evaluated offline, and that its output can drive concrete guest-facing actions in addition to informing pricing and inventory decisions.

G. Retrieval-Augmented AI Chatbots for Tourism and Reservation Assistance

Shih and Kang present the design and implementation of a secure retrieval-augmented-generation (RAG) chatbot for smart-tourism customer service, integrating API-based function calls and multi-layered linguistic analysis while defending against prompt-injection attacks [7]. The system is built to answer practical traveller queries and to route bookings and recommendations with heightened contextual awareness.

For a hotel reservation platform, this reference indicates that a conversational booking assistant should combine retrieval-based grounding in accurate, up-to-date property information with security safeguards, since a reservation chatbot that can be manipulated into leaking guest data or issuing unauthorised actions would pose a serious operational risk.

H. IoT-Based Smart-Hotel Safety and Room Management

Sudhakar, Saikrishnan, Arun Depak, and Ganesh propose an Internet-of-Things-based hardware and software solution for smart hotels that uses in-room sensors to detect hazards such as gas leaks, smoke, and fire, while a companion web application supports room booking, food and grocery ordering, and other guest services from a single platform [8]. The work addresses the practical problem of guests and staff having to rely on multiple disconnected applications for different hotel services.

This reference motivates extending the proposed reservation platform beyond the booking transaction itself into physical room management, so that occupancy sensing and safety monitoring can, in principle, feed back into room-status data used by the reservation and allocation logic.

I. Integrated IoT Context-Awareness Framework for Hotel Management

Hassan and Eassa present the Smart Hotel Management Information System (SHMIS), an architecture that integrates IoT context awareness with cloud computing to unify previously disconnected hotel subsystems and to improve both

guest experience and operational efficiency [9]. The framework responds to the observation that many traditional hotel systems lack interconnectivity, leading to resource wastage and delayed responses to changing conditions.

For the proposed platform, this work supports a layered architecture in which a cloud-based context-awareness layer aggregates room-sensor data, reservation data, and guest-preference data so that pricing, forecasting, and room-management modules can operate on a shared, real-time view of hotel state rather than isolated data silos.

J. Personalised Hotel Recommendation Systems

Balan, Popescu, and Mihaescu introduce a personalised hotel recommendation system that applies clustering techniques, such as k-means, to group similar hotels using guest ratings, hotel attributes, and contextual information, generating individualised recommendations without relying on extensive labelled training data [10]. Although the paper is not primarily a pricing or forecasting study, recommendation-system concepts are relevant to the guest-facing selection stage that typically precedes a reservation.

For the proposed reservation platform, this reference suggests that ranking candidate rooms or rate plans for a specific guest can draw on the same clustering and contextual-similarity techniques used in general hotel recommendation, complementing the price and demand signals produced by the forecasting components.

TABLE I
Comparative Analysis of the Reviewed Reference Papers

Ref.	Focus Area	AI / Technology Direction	Primary Relevance	Observed Scope	Limitation / Gap from Provided Citation
[1]	Dynamic pricing	Deep learning (DeepFM + seq2seq)	Occupancy forecasting linked to base-price setting	Hotel revenue management	Exact competitor-analysis dataset and full evaluation figures are not detailed in the abstract.
[2]	Revenue optimisation	Statistical data analysis (ANOVA, regression)	Identifying factors driving room price	Hotel pricing analytics	Predictive model architecture for real-time pricing is not specified.
[3]	Occupancy prediction	Price-elasticity modelling	Price-aware demand estimation	Room-level occupancy forecasting	Full dataset scale and deployment context require the complete paper.
[4]	Discriminatory pricing	Two-stage deep reinforcement learning	Fairness-constrained price optimisation	Hotel room pricing strategy	Real-world case-study scale and guest-response data are limited to the reported case study.
[5]	Cancellation forecasting	Bayesian probabilistic models	Uncertainty-aware cancellation prediction	Booking-cancellation analytics	Benchmark comparison against non-Bayesian baselines is limited in the available abstract.
[6]	Cancellation classification	Machine-learning classification, A/B testing	Net-demand calculation and guest outreach	Production hotel PMS deployment	Specific classifier hyperparameters are not detailed in the abstract.

Ref.	Focus Area	AI / Technology Direction	Primary Relevance	Observed Scope	Limitation / Gap from Provided Citation
[7]	Reservation chatbot	Retrieval-augmented generation, security guardrails	Secure conversational booking assistance	Smart-tourism customer service	Findings are based on a single regional case study (Hsinchu, Taiwan).
[8]	Room safety and services	IoT sensor network with web application	Hazard detection and unified guest services	Smart-hotel room management	Sensor accuracy and hardware cost figures are not stated in the abstract.
[9]	System integration	IoT context awareness with cloud computing	Unified, real-time hotel operations data	Smart hotel management architecture	Quantitative efficiency gains are not detailed in the abstract.
[10]	Recommendation	Unsupervised clustering (k-means)	Personalised hotel/room ranking	Guest-facing recommendation	Not a pricing or forecasting study; adaptation to reservation ranking is needed.

IV. COMPARATIVE ANALYSIS

The ten references collectively cover the principal stages required for an AI-based hotel reservation and room-management solution. Papers [1], [2], [3], and [4] emphasise dynamic pricing and occupancy-price interaction, with [1] and [3] focused on forecasting-driven pricing and [4] extending pricing toward fairness-constrained reinforcement learning. Papers [5] and [6] concentrate on cancellation prediction and its translation into net-demand estimates and operational outreach. Paper [7] contributes a conversational reservation interface, while [8] and [9] provide complementary IoT-based perspectives on physical room management and system-wide integration. Paper [10] contributes recommendation-system relevance for the guest-facing selection stage.

A clear pattern is the need to connect forecasting with pricing action. A demand or occupancy forecast by itself informs the system that conditions are changing, but a dynamic-pricing and room-management platform must transform that information into a concrete rate, an inventory decision, or a guest-facing recommendation. Therefore, the proposed platform should maintain a pipeline from data acquisition to preprocessing, demand and cancellation forecasting, price optimisation, reservation and room allocation, and continuous updating. The literature also indicates that fairness, security, and real-time responsiveness are important considerations, particularly for guest-facing chatbots and discriminatory pricing mechanisms [4], [7].

Because the reviewed references use different datasets, property types, and evaluation protocols, their reported approaches cannot be directly compared numerically from the information available here. A future experimental study should therefore establish a shared benchmark and evaluate candidate models using consistent measures such as occupancy-forecast error, cancellation-classification performance, revenue per available room, pricing fairness, recommendation relevance, and system latency.

V. LIMITATIONS OF EXISTING APPROACHES

First, the reviewed references address closely related but different parts of the hotel-management problem. Some emphasise pricing, some focus on cancellation and demand forecasting, and others address conversational booking or physical room management. This creates an opportunity for a single integrated platform that connects all stages.

Second, real-time hotel-booking environments are highly dynamic. A pricing or occupancy model trained on historical patterns may become less reliable when unusual events, local festivals, weather disruptions, or abrupt demand shocks occur. A practical platform therefore requires continuous data updates and mechanisms for adapting predictions and prices.

Third, dynamic and discriminatory pricing should account for guest-perceived fairness. If pricing is optimised purely for revenue without fairness constraints, it can generate negative guest sentiment and erode trust, as highlighted in the fairness-constrained pricing literature [4].

Fourth, conversational reservation assistants introduce security considerations. A chatbot that retrieves and acts on booking information must be protected against manipulation, such as prompt-injection attacks, that could expose guest data or trigger unauthorised reservation actions [7].

Finally, the reviewed sources do not provide enough information to establish a uniform comparison of datasets, model accuracy, or revenue-improvement percentages across property types and regions. These factors must be measured experimentally before claiming the superiority of one approach over another.

VI. RESEARCH GAPS

- Lack of a clearly unified pipeline combining real-time booking data acquisition, demand and cancellation forecasting, dynamic pricing, and room management within one end-to-end platform.
- Limited common evaluation methodology across the reviewed works for comparing pricing quality, forecasting accuracy, and guest-experience outcomes.

- Need for adaptive pricing and forecasting mechanisms that can respond to rapidly changing demand, cancellations, and unusual events.
- Need for fairness-aware and transparent pricing so that discriminatory or dynamic pricing does not simply maximise revenue at the expense of guest trust.
- Need for secure, retrieval-grounded conversational interfaces so that AI-driven reservation assistants remain accurate and resistant to manipulation.
- Need to connect physical room-management sensing (IoT) with reservation and pricing logic so that room status directly informs inventory decisions.
- Need for robust testing across different property types, room categories, seasons, and guest segments.

VII. PROPOSED DIRECTIONS FOR FUTURE WORK

Based on the reviewed literature, the proposed AI-Driven Hotel Reservation and Room Management Platform with Dynamic Pricing and Demand Forecasting should integrate booking-data acquisition, preprocessing, demand and cancellation forecasting, price optimisation, and reservation and room management in a single workflow. Real-time booking, occupancy, and competitor observations can be converted into features representing current and near-future demand conditions, after which a machine-learning or hybrid predictive model can estimate occupancy, cancellation probability, and net demand.

The pricing module can then evaluate candidate room rates using the predicted demand together with fairness constraints, as suggested in [4], so that revenue optimisation does not come at the expense of perceived pricing fairness. The conversational-reservation direction identified in [7] can be adopted for guest-facing booking assistance when a secure, retrieval-grounded interface is required, while the IoT-based room-management perspectives in [8] and [9] can guide the treatment of physical room status and safety within the same platform. The recommendation techniques discussed in [10] can additionally be specialised to rank candidate rooms or rate plans for individual guests.

Future implementation and evaluation should compare multiple pricing and forecasting models and establish a consistent experimental protocol. Evaluation should include both predictive performance and business-level outcomes, such as revenue per available room, occupancy improvement, cancellation-related loss reduction, guest-perceived fairness, and system responsiveness. The reservation interface should also provide guests with a clear explanation of the factors influencing a quoted room rate.

VIII. CONCLUSION

This literature survey reviewed ten reference works relevant to AI-driven hotel reservation and room management with dynamic pricing and demand forecasting. The references collectively demonstrate the growing use of artificial

intelligence for deep-learning-based pricing, price-elasticity and occupancy modelling, fairness-constrained discriminatory pricing, cancellation prediction, conversational reservation assistance, IoT-enabled room management, and personalised recommendation. The strongest common theme is the transition from static, manually managed hotel operations toward predictive and adaptive reservation and room-management systems [1]–[10].

The review also reveals an opportunity to integrate these capabilities into one end-to-end platform. An AI-Driven Hotel Reservation and Room Management Platform can use real-time booking and occupancy information to forecast demand and cancellations, convert those forecasts into fair and responsive room pricing, and extend intelligent management into the physical room through IoT sensing. The proposed research should focus on reliable forecasting, fairness-aware pricing, secure conversational interfaces, network-aware room management, and consistent evaluation so that the resulting platform is both technically effective and useful for real-world hotel operations.

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